Wunmin Jang

COMPREHENSIVE CURRICULUM VITAE | EDUCATION, HUMAN–AI INTERACTION & APPLIED RESEARCH

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PhD candidate (dissertation submitted; degree pending) | Technical University of Munich

ACADEMIC AND RESEARCH PROFILE

Interdisciplinary researcher working across learning sciences, educational psychology, human–AI interaction, computer-supported collaborative learning, and applied education and workforce-development research. Experienced in designing and analysing human-participant experiments, and in conducting supervised, project-based research on educational outcomes, instructor professional development, vocational quality assurance and digital-skills curricula. Research methods combine controlled interventions, pre/post assessments, gaze and interaction analytics, participant questionnaires, focus group interviews, survey analysis, and mixed-method interpretation. Conducted statistical analyses in Python for doctoral research and has experience with SPSS and AMOS in Korean educational research.

RESEARCH INTERESTS

AI-assisted learning and the distinction between performance and independent understanding; human–AI interaction and appropriate reliance; metacognition and cognitive autonomy (emerging research direction); cognitive workload and user experience; multimodal learning analytics; visual attention and eye tracking; collaborative problem solving and dialogue; teacher professional development and vocational education; learning outcomes, training transfer, survey measurement and psychometrics; education programme research and evaluation.

EDUCATION

PhD candidate, Social Sciences and Technology | Technical University of Munich (TUM), Germany | Jun 2023–present

- Dissertation: “Attention and communication dynamics in distributed pair programming and AI-assisted programming in computer science education.”
- Doctoral dissertation submitted September 2026; defence and degree award pending (expected early 2027, subject to university scheduling).

Master of Arts, International Office Administration | Ewha Womans University, Seoul, Republic of Korea | Mar 2021–Feb 2023

Bachelor’s degree, Civil Engineering | Kumoh National Institute of Technology, Republic of Korea | Mar 2015–Feb 2020

Exchange studies, Business Administration | Hochschule Heilbronn, Germany | Mar 2017–Feb 2018

DOCTORAL RESEARCH

Human–AI assistance, task performance and learning | TUM, Munich | 2023–2026

- Designed, conducted and analysed a controlled comparison of GitHub Copilot Chat and a curated Python tutorial with 64 recruited and 61 analysed university participants.
- Developed a main phase of 44 assisted Python programming tasks and unassisted pre/post assessments with eight items per test, separating performance during tool use from subsequent independent performance.
- Implemented research protocols, consent/ethics materials, participant scheduling and task delivery; assembled item-level code submissions, interface interaction records, eye-tracking recordings and survey responses.
- Evaluated task completion and rubric-scored correctness, gaze allocation and attention to assistance resources, subjective workload (NASA-TLX) and user experience (UEQ). The manuscript describes blinded human rubric scoring and reliability checks.
- Processed and analysed data in Python, including generalized estimating equations and mixed-effects modelling for repeated task- and participant-level observations.
- Observed markedly higher assisted task correctness in the Copilot condition, without a statistically detectable between-condition difference in short-term unassisted pre/post learning gains; analysed uncertainty and equivalence bounds.
- Conducted exploratory gender-related analyses of performance, learning and visual attention; no statistically reliable resource-by-gender interactions were found, and small/uneven subgroups limited inference.

Distributed pair programming, visual attention and team communication | TUM, Munich | 2023–2026

- Developed and conducted exploratory distributed pair-programming studies with 19 participants (10 experts and 9 students) in expert–expert, student–student and mixed-expertise dyads.
- Examined easy and difficult debugging tasks; collected screen-based eye-tracking data, verbal interaction data, focus-group interviews and questionnaires to characterise visual attention, communication and coordination.
- Defined areas of interest and analysed task-focused versus off-target gaze allocations; compared communication and shared-focus patterns across expertise composition and task difficulty.
- Interpreted differences cautiously as exploratory; published first-author results at ACM ETRA 2024 and IEEE ITET 2026.

Human versus LLM annotation of collaborative discourse | TUM, Munich | 2024

- Compared human coding with zero-shot LLM-based multi-label annotation of transactive communication in pair-programming transcripts, using a nine-category discourse scheme.
- Assessed precision, recall and F1, with the best tested LLM configuration achieving approximately 0.59 F1; examined the limitations of automated coding for interpreting educational dialogue.
- Published the findings as a first-author paper at IEEE ITHET 2024.

Review of machine-learning-integrated eye tracking in education | TUM / international coauthor collaboration | 2026

- Co-authored a systematic review of machine-learning approaches to gaze analysis in educational research, with attention to links between analytic outputs and educational or pedagogical decisions.
- Manuscript under review at ACM Transactions on Computing Education; not yet a published article.

APPLIED EDUCATION AND VOCATIONAL RESEARCH — REPUBLIC OF KOREA

The following team projects involved direct focus group participation, data analysis, report preparation and supervised schedule coordination, as applicable. Roles reflect project documentation and confirmed personal contributions; programme-wide outputs are credited to the research teams.

Quality management for vocational instructor continuing professional development | Ewha Womans University research collaboration | 2021–2022

- Participated directly in stakeholder focus-group interviews concerning the quality and delivery of in-service training for vocational training instructors.
- Contributed to qualitative evidence analysis, review of training processes, preparation of the final research report and coordination of research activities and milestones under professors' and senior doctoral researchers' supervision.
- Co-authored a peer-reviewed journal article on the team's ADDIE/PDCA-informed quality management framework, consisting of nine major stages and 61 detailed procedural elements.

Higher education learning experiences and outcomes by admission pathway | Ewha Womans University institutional research project | 2021–2022

- Served as a named co-researcher on a university study of students' learning experiences, learning satisfaction, educational outcomes and admission pathways.
- Participated in analysis and interpretation of institutional evidence, preparation of the final report, and schedule coordination/documentation under academic supervision.
- Contributed to a body of related co-authored publications on student experience, learning outcomes, university admission preferences and qualitative student experience.

Digital occupations, skills standards and vocational curricula | South Korean commissioned research project (Ewha research appointment) | 2022–2023

- Served as a named research assistant on a commissioned study of emerging digital occupations and corresponding competency requirements for vocational education and training.
- Participated directly in FGI-related research, occupational and skills analysis, interpretation of findings, final-report drafting and supervised research scheduling.
- Contributed to team outputs describing occupational roles, common competencies and standard training programmes, including web programming and 3D modelling.

Other collaborative educational and organisational research | South Korean academic collaborations | 2022–2024

- Co-authored peer-reviewed studies of instructor training transfer using the Success Case Method, psychometric validation of the Korean Career Crafting Assessment, career capital research trends, and organisational behaviour involving empowering leadership and positive psychological capital.

- Participated in survey-based and mixed-method educational research; used or contributed to analyses involving SPSS, AMOS, regression, ANOVA, chi-square tests, structural equation modelling and instrument validation.

RESEARCH AND TECHNICAL COMPETENCIES

Research design and evidence collection

Controlled human-participant experiments; between-group comparisons; task and assessment design; pre/post learning measures; student performance assessment; rubric-based scoring; ethics and consent documentation; participant recruitment/scheduling; questionnaires and survey research; interviews and focus group interviews; qualitative and quantitative evidence synthesis; research documentation and data-quality checks.

Human behaviour, cognition and collaboration

Human–AI interaction; cognitive offloading and reliance questions; NASA Task Load Index (NASA-TLX); User Experience Questionnaire (UEQ); distributed pair programming; collaborative problem-solving; transactive communication and discourse coding; human versus LLM annotation comparison; basic metacognitive and learning-outcome evaluation.

Multimodal analytics and software

Tobii eye tracking and Tobii Pro Lab; area-of-interest (AOI) definition and gaze-allocation analysis; visual attention measures; interface/interaction logs; verbal interaction transcripts; Python; SPSS; AMOS; statistical visualisation and reporting.

Statistical and methodological approaches

Regression, analysis of variance (ANOVA), chi-square tests, structural equation modelling (SEM), generalized estimating equations (GEE), mixed-effects models/GLMM, repeated-measures comparisons, task-level outcome modelling, survey instrument validation, qualitative coding and mixed-method triangulation. Specific methods vary by project; not all were used in every study.

Project coordination and communication

Research milestone scheduling; coordination of participant sessions and data collection; supervised coordination of applied research tasks; collaboration with professors and doctoral researchers; focus-group participation; final-report and manuscript drafting; presentation of research findings; communication of evidence through tables, figures, academic papers and written reports.

AWARDS, HONOURS AND RESEARCH RECOGNITION

- Best Presentation Award — 7th International Conference on Information Technology and Education Technology (ITET 2026), for first-author work on visual attention and communication in distributed pair programming.
- Laura Bassi Scholarship — November 2024 to October 2025.
- Honorable Mention — Study on Validation of the Korean Career Crafting Assessment (K-CCA), Korea Tech Employment & Vocational Competency Development Center, 2022.

PEER-REVIEWED PUBLICATIONS

13 published peer-reviewed works: three international conference papers and ten Korean-language journal articles. Titles of Korean-language articles are given in English for international use.

International conference papers (3)

01. Jang, W., Gao, H., Michaeli, T., & Kasneci, E. (2026). **Understanding Visual Attention and Communication in Distributed Pair Programming for Programming Education.** 7th International Conference on Information Technology and Education Technology (ITET), pp. 101–106. DOI: 10.1109/ITET70052.2026.11650705. Best Presentation Award.

02. Jang, W., Hou, R., Gao, H., & Kasneci, E. (2024). **Analyzing Communication Logs in Pair Programming: A Comparison of Human- and LLM-Based Approaches.** 21st International Conference on Information Technology Based Higher Education and Training (ITHET). DOI: 10.1109/ITHET61869.2024.10837662.

03. Jang, W., Gao, H., Michaeli, T., & Kasneci, E. (2024). **Exploring Communication Dynamics: Eye-tracking Analysis in Pair Programming of Computer Science Education.** Proceedings of the ACM Symposium on Eye Tracking Research and Applications (ETRA), Article 65. DOI: 10.1145/3649902.3653942.

Korean-language journal articles (10)

04. Park, Y., Yim, S., Jang, W., Kim, J., & Song, Y. (2024). **Analysis of Factors Influencing University Admission Type Preferences: The Case of A University.** The Journal of Learner-Centered Curriculum and Instruction, 24(11), 523–542. DOI: 10.22251/jlcci.2024.24.11.523.

05. Lee, J. Y., Park, Y., Nam, Y., Jang, W., Kim, J., & Song, Y. (2023). **Analysis of Entry-Level 3D Modeler Job and Development of a Standard Occupational Training Program for Career Technical Education Students.** The Journal of Learner-Centered Curriculum and Instruction, 23(24), 337–351. DOI: 10.22251/jlcci.2023.23.24.337.

06. Park, Y., Yim, S., Jang, W., Kim, J., & Song, Y. (2023). **The Structural Relationship between College Students' Personal Characteristics, Learning Experiences, Learning Satisfaction, and Learning Outcomes.** The Journal of Learner-Centered Curriculum and Instruction, 23(21), 495–517. DOI: 10.22251/jlcci.2023.23.21.495.

07. Park, Y., Lee, J. Y., Nam, Y., Jang, W., Kim, J., & Song, Y. (2023). **Job Analysis and Training Course Development of Web Programmer for Students at Specialized Vocational High Schools.** The Journal of Vocational Education Research, 42(4), 73–94.

08. Park, Y., Yim, S., Jang, W., Kim, J., & Song, Y. (2023). **A Qualitative Study on the Learning Experiences of College Students Who Cross-Applied from Humanities to Natural Sciences: The Case of University A.** The Journal of Educational Studies, 54(2), 51–78. DOI: 10.15854/jes.2023.06.54.2.51.

09. Park, Y., Lee, H., Jang, W., & Kim, J. (2023). **An Analysis on the Research Trends of Career Capital: Focused on Journals Published in Korea and Abroad.** Journal of Korea Academia-Industrial Cooperation Society, 24(5), 471–489. DOI: 10.5762/KAIS.2023.24.5.471.

10. Jang, W., Lee, J. Y., & Park, Y. (2023). **The Influences of Leader's Empowering Leadership on In-Role Behavior and Extra-Role Behavior of Members of the MZ Generation in Public Institutions: The Mediating Effect of Positive Psychological Capital.** Journal of Corporate Education and Talent Research, 25(1), 141–178. DOI: 10.46260/KSLP.25.1.6.

11. Park, Y., Lee, S., Yim, S., Kim, Jiyoung, Jang, W., Kim, Jisoo, & Song, Y. (2022). **Analyzing Success Factors of Transfer of Training in In-Service Training for Vocational Training Instructors Using Success Case Method.** Journal of Vocational Education & Training, 25(3), 185–217. DOI: 10.36907/krivet.2022.25.3.185.

12. Park, Y., Lee, S., Yim, S., Kim, Jiyoung, Jang, W., Song, Y., & Kim, Jisoo. (2022). **Development of Quality Management System for In-Service Training for Vocational Training Instructors.** Journal of Vocational Education & Training, 25(2), 243–277. DOI: 10.36907/krivet.2022.25.2.243.

13. Yim, S., Jang, W., Song, Y., & Park, Y. (2022). **A Study on the Validation of the Korean Version of the Career Crafting Assessment (K-CCA).** Journal of Corporate Education and Talent Research, 24(4), 111–146. DOI: 10.46260/KSLP.24.4.4.

MANUSCRIPTS UNDER REVIEW

14. Jang, W., Özdel, S., Abdrabou, Y., Gao, H., & Kasneci, E. (2026). **Task Performance, Short-Term Learning, and Visual Attention in AI-Assisted Programming with GitHub Copilot.** Manuscript under review at IEEE Transactions on Learning Technologies.

15. Jang, W., Abdrabou, Y., Molnar, A., & Kasneci, E. (2026). **From Gaze Analytics to Pedagogical Action: A Systematic Review of Machine Learning–Integrated Eye Tracking in Education with Implications for Computing Education.** Manuscript under review at ACM Transactions on Computing Education.

LANGUAGES AND PROFESSIONAL CONNECTIONS

Korean — Native | English — Fluent (C1) | German — Intermediate (B1)